

MATH 241

Problem Set - 1

Question 1.

Let $z = a + jb$

$$\begin{aligned} |\operatorname{Im}\{1 - 3\bar{z} + z^2\}| &= |\operatorname{Im}\{1 - 3(a - jb) + (a^2 - b^2) + j2ab\}| \\ &= |3b + 2ab| \leq 3|b| + 2|a||b| \end{aligned}$$

Thus $|z| = \sqrt{a^2 + b^2} < 1 \Rightarrow |a|, |b| < 1$

$$\Rightarrow 3|b| + 2|a||b| < 5$$

$$\Rightarrow |3b + 2ab| < 5$$

$$\Rightarrow |\operatorname{Im}\{1 - 3\bar{z} + z^2\}| < 5$$

Question 2.

Let $z = a + jb$

$$(|a| - |b|)^2 \geq 0$$

$$a^2 + b^2 - 2|a||b| \geq 0$$

$$2(a^2 + b^2) - a^2 - b^2 - 2|a||b| \geq 0$$

$$2(a^2 + b^2) \geq a^2 + 2|a||b| + b^2$$

$$2(a^2 + b^2) \geq (|a| + |b|)^2$$

$$\sqrt{2} \sqrt{a^2 + b^2} \geq |a| + |b|$$

$$\sqrt{2} |z| \geq |\operatorname{Re}(z)| + |\operatorname{Im}(z)|$$

Question 3.

Let $z = b + jc$, $b, c \in \mathbb{R}$

$$|z + a|^2 - 2\operatorname{Re}\{az\} = |(a + b) + jc|^2 - 2\operatorname{Re}\{ab + jac\}$$

$$= a^2 + 2ab + b^2 + c^2 - 2ab$$

$$= (b^2 + c^2) + a^2$$

$$= |z|^2 + a^2$$

Question 4.

$$\text{Let } z = x_1 + jy_1, \quad b = x_2 + jy_2$$

$$\begin{aligned} |z + \bar{b}|^2 - |b|^2 &= |(x_1 + x_2) + j(y_1 - y_2)|^2 - |x_2 + jy_2|^2 \\ &= x_1^2 + 2x_1x_2 + \cancel{x_2^2} + y_1^2 - 2y_1y_2 + \cancel{y_2^2} - \cancel{x_2^2} - \cancel{y_2^2} \\ &= x_1^2 + y_1^2 + 2(x_1x_2 - y_1y_2) \\ &= |z|^2 + 2 \operatorname{Re}\{bz\} \end{aligned}$$

Question 5. (A)

We know the triangle inequality: $|x+y| \leq |x|+|y|$

$$\begin{aligned} \text{Let } x=w, y=z-w &\Rightarrow |z| \leq |w|+|z-w| \Rightarrow |z-w| \geq |z|-|w| \\ \text{Let } x=z, y=w-z &\Rightarrow |w| \leq |z|+|w-z| \Rightarrow |z-w| \geq |w|-|z| \end{aligned} \quad \left. \vphantom{\begin{aligned} \text{Let } x=w, y=z-w \\ \text{Let } x=z, y=w-z \end{aligned}} \right\} \Rightarrow |z-w| \geq ||z|-|w||$$

(B)

$$|z|=3$$

$$\frac{1}{|z^4 - 7z^2 + 10|} = \frac{1}{|z^2 - 5| \cdot |z^2 - 2|} \leq \frac{1}{||z^2|-5| \cdot ||z^2|-2|} = \frac{1}{|9-5| \cdot |9-2|} = \frac{1}{28}$$

Question 6.

From 5.(A) we have

$$\begin{aligned} |p(z)| &\geq |a_n||z|^n - |a_0 + a_1z + \dots + a_{n-1}z^{n-1}| \\ &\geq |a_n||z|^n - (|a_0| + |a_1||z| + \dots + |a_{n-1}||z|^{n-1}) \\ &\geq |a_n||z|^n - (|a_0| + |a_1| + \dots + |a_{n-1}|)|z|^{n-1} \quad \leftarrow |z| \geq 1, \text{ so we can multiply each term to make them larger, and the overall expression smaller} \\ &\geq |a_n||z|^n - (|a_0| + |a_1| + \dots + |a_{n-1}|)|z|^n \frac{1}{|z|} \quad \leftarrow |z| \geq 2 \frac{|a_0| + |a_1| + \dots + |a_{n-1}|}{|a_n|} \\ &\geq |a_n||z|^n - (|a_0| + |a_1| + \dots + |a_{n-1}|)|z|^n \frac{|a_n|}{2(|a_0| + |a_1| + \dots + |a_{n-1}|)} \\ &\geq \frac{1}{2}|a_n||z|^n \end{aligned}$$

Question 7.

Let $y_1 = \alpha_1 + k\beta_1$, $y_2 = \alpha_2 + k\beta_2$, $y_3 = \alpha_3 + k\beta_3$

Define addition as $y_1 + y_2 = (\alpha_1 + \alpha_2) + k(\beta_1 + \beta_2)$

Define multiplication as $y_1 \cdot y_2 = (\alpha_1\alpha_2 - \beta_1\beta_2) + k(\alpha_1\beta_2 + \beta_1\alpha_2 - \beta_1\beta_2)$

This definition can be motivated by:

$$(\alpha_1 + k\beta_1) \cdot (\alpha_2 + k\beta_2) = \alpha_1\alpha_2 + k\alpha_1\beta_2 + k\beta_1\alpha_2 + k^2\beta_1\beta_2$$

and since $k^2 + k + 1 = 0 \Rightarrow k^2 = -k - 1$,

$$(\alpha_1 + k\beta_1) \cdot (\alpha_2 + k\beta_2) = \alpha_1\alpha_2 + k\alpha_1\beta_2 + k\beta_1\alpha_2 + (-k-1)\beta_1\beta_2$$

Note that for $\beta_1 = \beta_2 = 0$, these definitions reduce to the ordinary rules.

Proof that K is a field:

- $y_1 + y_2 = (\alpha_1 + \alpha_2) + k(\beta_1 + \beta_2) = (\alpha_2 + \alpha_1) + k(\beta_2 + \beta_1) = y_2 + y_1$

- $y_1 + (y_2 + y_3) = [\alpha_1 + (\alpha_2 + \alpha_3)] + k[\beta_1 + (\beta_2 + \beta_3)]$
 $= [(\alpha_1 + \alpha_2) + \alpha_3] + k[(\beta_1 + \beta_2) + \beta_3] = (y_1 + y_2) + y_3$

- $0 = 0 + k0$ is the additive identity, since $0 + y_1 = (0 + \alpha_1) + k(0 + \beta_1) = \alpha_1 + k\beta_1 = y_1$

- $(-\alpha_1) + k(-\beta_1)$ is the additive inverse of y_1 , since $(-\alpha_1) + k(-\beta_1) + y_1 = 0 + k0 = 0$

- $y_1 \cdot y_2 = (\alpha_1\alpha_2 - \beta_1\beta_2) + k(\alpha_1\beta_2 + \beta_1\alpha_2 - \beta_1\beta_2)$
 $= (\alpha_2\alpha_1 - \beta_2\beta_1) + k(\alpha_2\beta_1 + \beta_2\alpha_1 - \beta_2\beta_1) = y_2 \cdot y_1$

- $y_1 \cdot (y_2 \cdot y_3) = (\alpha_1 + k\beta_1) [(\alpha_2\alpha_3 - \beta_2\beta_3) + k(\alpha_2\beta_3 + \beta_2\alpha_3 - \beta_2\beta_3)]$
 $= (\alpha_1\alpha_2\alpha_3 - \alpha_1\beta_2\beta_3 - \beta_1\alpha_2\beta_3 - \beta_1\beta_2\alpha_3 + \beta_1\beta_2\beta_3)$
 $+ k(\beta_1\alpha_2\alpha_3 - \beta_1\beta_2\beta_3 + \alpha_1\alpha_2\beta_3 + \alpha_1\beta_2\alpha_3 - \alpha_1\beta_2\beta_3 - \beta_1\alpha_2\beta_3 - \beta_1\beta_2\alpha_3 + \beta_1\beta_2\beta_3)$
 $= [(\alpha_1\alpha_2 - \beta_1\beta_2) + k(\alpha_1\beta_2 + \beta_1\alpha_2 - \beta_1\beta_2)] (\alpha_3 + k\beta_3) = (y_1 \cdot y_2) \cdot y_3$

- $y_1 \cdot (y_2 + y_3) = (\alpha_1 + k\beta_1) [(\alpha_2 + \alpha_3) + k(\beta_2 + \beta_3)]$
 $= (\alpha_1\alpha_2 + \alpha_1\alpha_3 - \beta_1\beta_2 - \beta_1\beta_3) + k(\alpha_1\beta_2 + \alpha_1\beta_3 + \beta_1\alpha_2 + \beta_1\alpha_3 - \beta_1\beta_2 - \beta_1\beta_3)$
 $= (\alpha_1\alpha_2 - \beta_1\beta_2) + k(\alpha_1\beta_2 + \beta_1\alpha_2 - \beta_1\beta_2) + (\alpha_1\alpha_3 - \beta_1\beta_3) + k(\alpha_1\beta_3 + \beta_1\alpha_3 - \beta_1\beta_3) = (y_1 \cdot y_2) + (y_1 \cdot y_3)$

- $1 = 1 + k0$ is the multiplicative identity, since $1 \cdot y_1 = (1\alpha_1 - 0\beta_1) + k(1\beta_1 + 0\alpha_1 - 0\beta_1) = \alpha_1 + k\beta_1 = y_1$

- For $y_1 \neq 0$, a candidate for multiplicative inverse can be founded by

$$\begin{aligned} \frac{1}{\alpha_1 + k\beta_1} &= \frac{\alpha_1 - \beta_1 - k\beta_1}{(\alpha_1 + k\beta_1)(\alpha_1 - \beta_1 - k\beta_1)} && \text{To cancel the imaginary part of the denominator with our new multiplication definition} \\ &= \frac{\alpha_1 - \beta_1 - k\beta_1}{\alpha_1^2 - \alpha_1\beta_1 + \beta_1^2 + k(\cancel{\alpha_1\beta_1} - \beta_1^2 - \cancel{\alpha_1\beta_1} + \beta_1^2)} \\ &= \frac{\alpha_1 - \beta_1}{\alpha_1^2 - \alpha_1\beta_1 + \beta_1^2} - k \frac{\beta_1}{\alpha_1^2 - \alpha_1\beta_1 + \beta_1^2} \end{aligned}$$

It can be shown that this is indeed the multiplicative inverse for $y_1 \neq 0$:

$$\begin{aligned} (\alpha_1 + k\beta_1) \left(\frac{\alpha_1 - \beta_1}{\alpha_1^2 - \alpha_1\beta_1 + \beta_1^2} - k \frac{\beta_1}{\alpha_1^2 - \alpha_1\beta_1 + \beta_1^2} \right) \\ &= \frac{1}{\alpha_1^2 - \alpha_1\beta_1 + \beta_1^2} \left[\alpha_1^2 - \alpha_1\beta_1 + \beta_1^2 + k(\cancel{\alpha_1\beta_1} - \beta_1^2 - \cancel{\alpha_1\beta_1} + \beta_1^2) \right] \\ &= \frac{\alpha_1^2 - \alpha_1\beta_1 + \beta_1^2}{\alpha_1^2 - \alpha_1\beta_1 + \beta_1^2} = 1 \end{aligned}$$

This is true for all $y_1 \neq 0$, because the denominator $\alpha_1^2 - \alpha_1\beta_1 + \beta_1^2 = (\alpha_1 - \frac{\beta_1}{2})^2 + \frac{3}{4}\beta_1^2$ is 0 if and only if $\alpha_1 = \beta_1 = 0$.

In this case, the multiplicative identity and multiplicative inverse was easy to guess; in general, they can be found using a system of linear equations, as shown in the following pages.

Q7 Let $u, v \in K$. Then $u = \alpha_1 + \beta_1 k$
 $v = \alpha_2 + \beta_2 k$

$$u+v = (\alpha_1 + \alpha_2) + k(\beta_1 + \beta_2)$$

$$uv = (\alpha_1 \alpha_2 + \underbrace{\beta_1 \beta_2 k^2}_{\substack{= \\ -k-1}} + (\alpha_1 \beta_2 + \alpha_2 \beta_1) k$$

$$= u \cdot v = \alpha_1 \alpha_2 - \beta_1 \beta_2 + k(\alpha_1 \beta_2 + \alpha_2 \beta_1 - \beta_1 \beta_2)$$

First 4 field properties are automatically satisfied
 (additive identity is 0)

Commutativity of multiplication is also clear -

~~is there a multiplicative identity?~~

Does there exist multiplicative identity?

We want to find $v = \alpha_2 + \beta_2 k$ s.t. for
 any $u = \alpha_1 + \beta_1 k \neq 0$, we have $uv = u$, i.e.

$$\underbrace{\alpha_1 \alpha_2 - \beta_1 \beta_2 + k(\alpha_1 \beta_2 + \alpha_2 \beta_1 - \beta_1 \beta_2)}_{\uparrow} = \underbrace{\alpha_1 + \beta_1 k}_{\downarrow}$$

$$\left. \begin{aligned} \alpha_1 \alpha_2 - \beta_1 \beta_2 &= \alpha_1 \\ \alpha_1 \beta_2 + \alpha_2 \beta_1 - \beta_1 \beta_2 &= \beta_1 \end{aligned} \right\} \begin{array}{l} \text{A linear system with 2 eqns} \\ \text{in 2 unknowns } \alpha_2, \beta_2 \end{array}$$

$$\begin{bmatrix} \alpha_1 & -\beta_1 \\ \beta_1 & \alpha_1 - \beta_1 \end{bmatrix} \begin{bmatrix} \alpha_2 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \beta_1 \end{bmatrix}$$

this system has a unique soln iff $\begin{vmatrix} \alpha_1 & -\beta_1 \\ \beta_1 & \alpha_1 - \beta_1 \end{vmatrix} \neq 0$

i.e. $\alpha_1^2 + \beta_1^2 - \alpha_1 \beta_1 \neq 0$

Observe that $\alpha_1^2 + \beta_1^2 - \alpha_1 \beta_1 = 0$ if & only if $\alpha_1 = \beta_1 = 0$ which is not the case.

Hence,
$$\begin{bmatrix} \alpha_2 \\ \beta_2 \end{bmatrix} = \frac{1}{\alpha_1^2 + \beta_1^2 - \alpha_1 \beta_1} \begin{bmatrix} \alpha_1 - \beta_1 & \beta_1 \\ -\beta_1 & \alpha_1 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \beta_1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\alpha_1^2 - \alpha_1 \beta_1 + \beta_1^2}{\alpha_1^2 + \beta_1^2 - \alpha_1 \beta_1} \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

That is, $u = 1$ is the multiplicative identity.

For the

* Multiplicative inverse: For $u \in K$ we need

to find $v = \alpha_2 + K\beta_2$ s.t. $u \cdot v = 1$.

i.e.
$$\begin{bmatrix} \alpha_1 & -\beta_1 \\ \beta_1 & \alpha_1 - \beta_1 \end{bmatrix} \begin{bmatrix} \alpha_2 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \alpha_2 \\ \beta_2 \end{bmatrix} = \frac{1}{\alpha_1^2 + \beta_1^2 - \alpha_1 \beta_1} \begin{bmatrix} \alpha_1 - \beta_1 & \beta_1 \\ -\beta_1 & \alpha_1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\alpha_1 - \beta_1}{\alpha_1^2 + \beta_1^2 - \alpha_1 \beta_1} \\ -\frac{\beta_1}{\alpha_1^2 + \beta_1^2 - \alpha_1 \beta_1} \end{bmatrix}$$

i.e. multiplicative inverse of $u = \alpha_1 + k\beta_1$ is

$$v = \left[\frac{\alpha_1 - \beta_1}{\alpha_1^2 + \beta_1^2 - \alpha_1\beta_1} \right] + k \left[\frac{-\beta_1}{\alpha_1^2 + \beta_1^2 - \alpha_1\beta_1} \right]. \text{ Indeed,}$$

$$u.v = \frac{\alpha_1^2 - \alpha_1\beta_1 \overset{k+1}{\overset{11}{-k^2}} \beta_1^2 + k(\alpha_1\beta_1 - \beta_1^2 - \alpha_1\beta_1)}{\alpha_1^2 + \beta_1^2 - \alpha_1\beta_1}$$

$$= \frac{\alpha_1^2 + \beta_1^2 - \alpha_1\beta_1 + \cancel{k(\beta_1^2 - \beta_1^2)}}{\alpha_1^2 + \beta_1^2 - \alpha_1\beta_1} = 1 //$$